

Loan Approval Model Evaluation Report

This report presents the performance comparison of four Machine Learning classification models using the 70K Loan Approval dataset.

Dataset Information

Information	Value
Training Samples	55994
Testing Samples	13999
Features	22
Models Compared	4

Best Performing Model

Metric	Score
Model	XGBoost
CV F1	0.9487
Train F1	0.9703
Accuracy	0.9490
Precision	0.9445
Recall	0.9540
Test F1	0.9492
Overfitting Gap	0.0211

Model Comparison

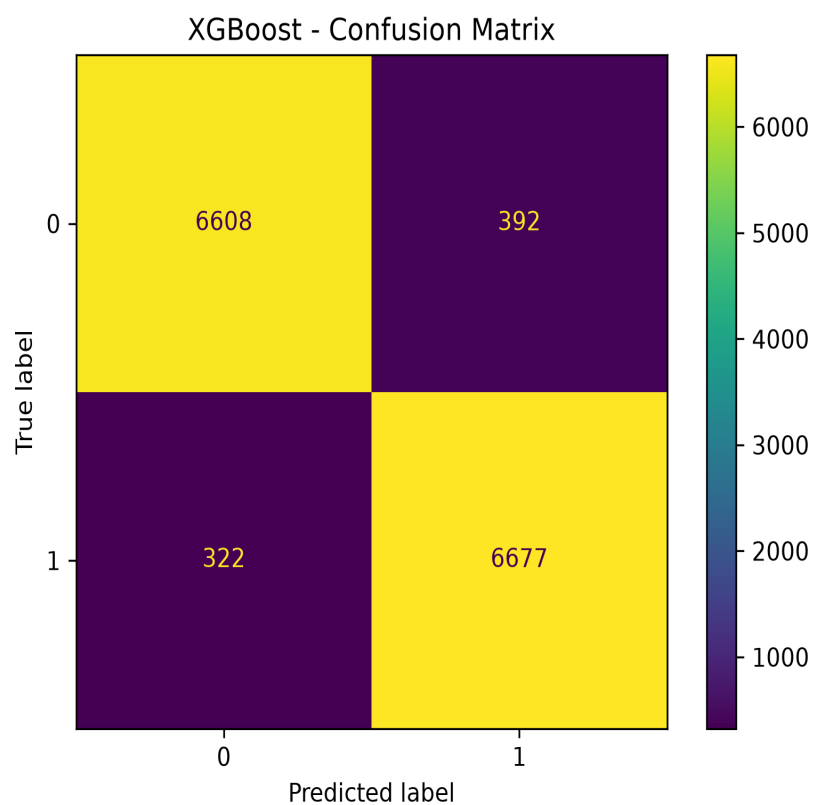
Model	CV F1	Train F1	Accuracy	Precision	Recall	F1 Score	Overfitting Gap
XGBoost	0.9487	0.9703	0.9490	0.9445	0.9540	0.9492	0.0211
Random Forest	0.9427	1.0000	0.9466	0.9426	0.9510	0.9468	0.0532
Decision Tree	0.9165	1.0000	0.9179	0.9144	0.9221	0.9183	0.0817

Logistic Regression	0.9144	0.9145	0.9156	0.9019	0.9327	0.9170	-0.0025
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XGBoost

Metric	Score
CV F1	0.9487
Train F1	0.9703
Accuracy	0.9490
Precision	0.9445
Recall	0.9540
Test F1	0.9492
Overfitting Gap	0.0211

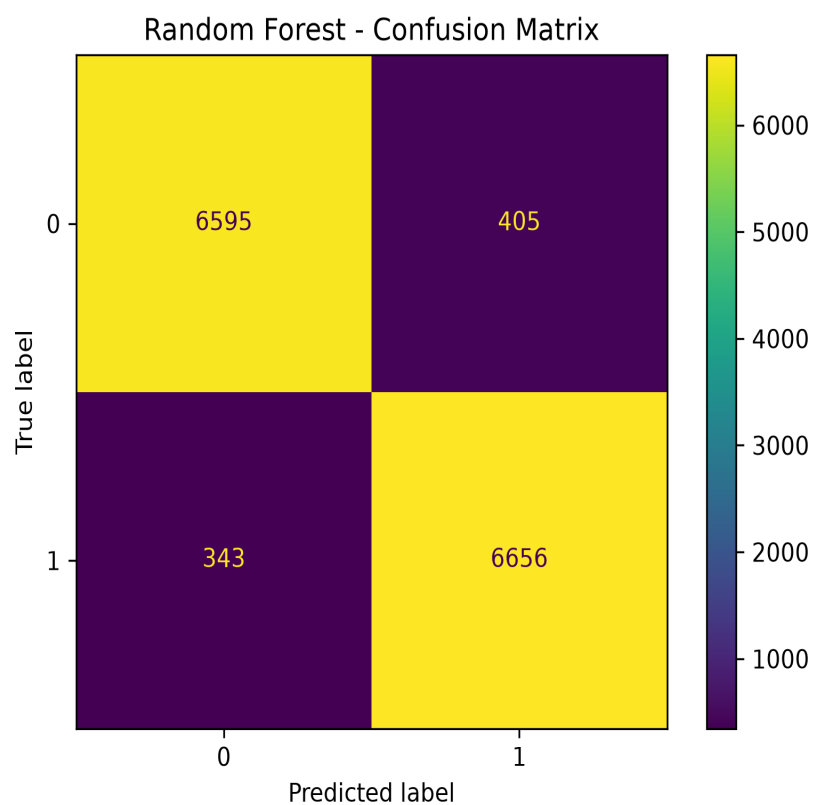
Confusion Matrix



Random Forest

Metric	Score
CV F1	0.9427
Train F1	1.0000
Accuracy	0.9466
Precision	0.9426
Recall	0.9510
Test F1	0.9468
Overfitting Gap	0.0532

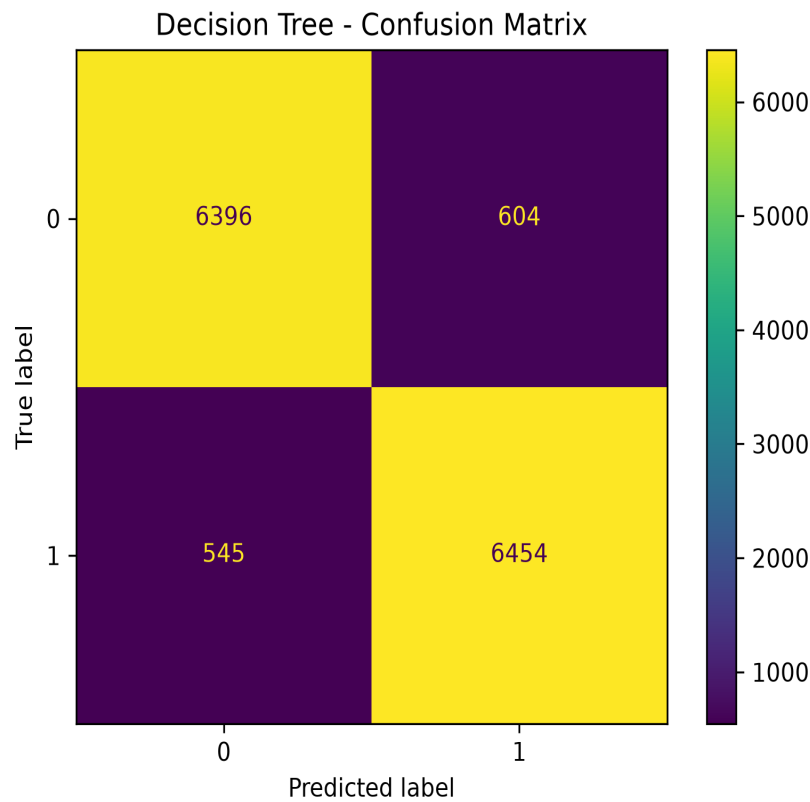
Confusion Matrix



Decision Tree

Metric	Score
CV F1	0.9165
Train F1	1.0000
Accuracy	0.9179
Precision	0.9144
Recall	0.9221
Test F1	0.9183
Overfitting Gap	0.0817

Confusion Matrix



Logistic Regression

Metric	Score
CV F1	0.9144
Train F1	0.9145
Accuracy	0.9156
Precision	0.9019
Recall	0.9327
Test F1	0.9170
Overfitting Gap	-0.0025

Confusion Matrix

